

■ AEM Bank Website

Complete Developer Study Guide

Sessions 1–22 + Hands-On Project

Sessions Covered	Sessions 1–22 (Complete Syllabus)
Project	AEM Bank Website — Maven Archetype 56
AEM Version	AEM 6.5
Java Version	Java 11 (Oracle JDK)
Maven Version	Maven 3.9.15
Key Flag	-Dhttps.protocols=TLSv1.2 (required on corporate network)
Project Path	C:\aemwork\laembank
Deploy Command	mvn clean install -Dhttps.protocols=TLSv1.2 -DskipTests -pl !ui.tests -PautoInstallPackage

1. Syllabus Session Map

Session	Topic	Key Concepts
Sessions 1–2	Foundation	CMS intro, AEM platform, Maven project setup, VS Code, deploy
Sessions 3–4	Templates & Components	Authoring, Layout Containers, Policies, Custom component structure
Sessions 5–6	Dialog & HTL	Touch UI dialog, Granite UI, HTL (Sightly) basics and advanced
Sessions 7–8	Sling Models & Templates	Sling Models, @ValueMapValue, @PostConstruct, editable templates, clientlibs
Sessions 9–10	Fragments	Experience Fragments, Content Fragments, Content Fragment Models
Sessions 11–12	Advanced UI	Dialog listeners, validators, conditional display, RTE plugins
Sessions 13–14	OSGi	OSGi components, services, lifecycle, @Component, @Reference, config nodes
Session 15	Sling Servlets	Path-based and resource-type servlets, @SlingServlet
Session 16	Security	Users, Groups, Permissions, ACLs, Impersonation
Session 17	Service Users & Logging	Service users, repointit, custom log files, runmodes
Session 18	Tagging	Tagging framework, TagManager API, tag search, QueryBuilder
Session 19	AEM Assets	DAM, metadata schemas, Asset API, AssetManager, workflows
Session 20	Workflows	Workflow models, WorkflowProcess, WorkItem, participant steps
Session 21	JUnit & Best Practices	AEM Mocks, Mockito, @Mock, @InjectMocks, coding standards, ACS Commons
Session 22	Replication & MSM	Replication agents, Author→Publish, reverse replication, flush agent, Live Copy, Blueprints

2. Project Setup — AEM Bank Website

2.1 Generate Project

Run from C:\aemwork using the temp pom.xml trick:

```
Set-Content -Path C:\aemwork\pom.xml -Value '<project><modelVersion>4.0.0</modelVersion><groupId>temp</groupId><artifactId>temp</artifactId><version>1</version><packaging>pom</packaging></project>'
```

```
mvn "-Dhttps.protocols=TLSv1.2" archetype:generate -B -DarchetypeGroupId=com.adobe.aem
-DarchetypeArtifactId=aem-project-archetype -DarchetypeVersion=56 -DaemVersion=6.5.22
-DartifactId=aembank -DappTitle="AEM Bank Website" -DappId=aembank -DgroupId=com.aembank
-DincludeDispatcherConfig=n -DfrontendModule=none
```

2.2 Standard Deploy Command

```
cd C:\aemwork\aembank mvn clean install "-Dhttps.protocols=TLSv1.2" -DskipTests "-pl"
"!ui.tests" -PautoInstallPackage
```

Key insight: -Dhttps.protocols=TLSv1.2 is required on corporate networks to fix SSL handshake failures. Always include this flag. -pl !ui.tests skips the browser automation tests that need Node.js.

2.3 Project Structure

Folder	Purpose
core/src/main/java/com/aembank/core/models/	Sling Models (.java)
core/src/main/java/com/aembank/core/services/	OSGi Services (.java)
core/src/main/java/com/aembank/core/servlets/	Sling Servlets (.java)
core/src/main/java/com/aembank/core/workflows/	Workflow process steps (.java)
core/src/test/java/com/aembank/core/models/	JUnit test classes
ui.apps/.../apps/aembank/components/	HTL components (.html + .content.xml)
ui.apps/.../apps/aembank/clientlibs/	Client libraries (CSS/JS)
ui.config/.../apps/aembank/osgiconfig/	OSGi config XML files
ui.content/.../content/aembank/	Site pages

3. Key Concepts Per Session

Session 7 — Sling Models

A Sling Model reads JCR properties saved by a dialog and exposes them to HTL via getters.

```
@Model(adaptables=SlingHttpServletRequest.class,
defaultInjectionStrategy=DefaultInjectionStrategy.OPTIONAL) public class HeroBannerModel {
@ValueMapValue private String bannerTitle; // reads 'bannerTitle' JCR property @ValueMapValue
private String ctaLink; private boolean hasCtaButton; @PostConstruct protected void init() {
// compute derived values here – runs ONCE after injection hasCtaButton = ctaText != null &&
ctaLink != null; } public String getBannerTitle() { return bannerTitle; } // HTL:
${model.bannerTitle} public boolean isHasCtaButton() { return hasCtaButton; } // HTL:
${model.hasCtaButton} }
```

Rules: Always use OPTIONAL strategy. All logic in @PostConstruct, never in getters. Field name must match JCR property name saved by dialog.

Session 13 — OSGi Services

OSGi services are Java singletons managed by the OSGi container. Use @Component to register, @Reference to inject.

```
@Component(service=BankConfigService.class) public class BankConfigServiceImpl implements
BankConfigService { @Activate protected void activate(BankConfig config) { this.apiUrl =
config.apiUrl(); } @ObjectClassDefinition(name="Bank Config") @interface BankConfig { String
apiUrl() default "https://api.bank.com"; } }
```

Session 17 — Service Users, Logging, Runmodes

Service users allow background jobs to access JCR without using admin credentials.

```
// Get resolver using service user (not admin!) Map auth = Collections.singletonMap(
ResourceResolverFactory.SUBSERVICE, "bankwebsite-subservice"); try (ResourceResolver resolver
= resolverFactory.getServiceResourceResolver(auth)) { Resource r =
resolver.getResource("/content/aembank"); }
```

Runmodes: config/ = all instances | config.author/ = author only | config.author.dev/ = author + dev only

Session 20 — Workflows

Custom workflow process steps implement WorkflowProcess and are picked up by AEM's workflow engine.

```
@Component(service=WorkflowProcess.class, property={"process.label=Bank: Validate and
Publish"}) public class ValidateAndPublishProcess implements WorkflowProcess { public void
execute(WorkItem item, WorkflowSession wfSession, MetadataMap args) throws WorkflowException {
String path = item.getWorkflowData().getPayload().toString(); // validate, then replicate
replicator.replicate(session, ReplicationActionType.ACTIVATE, path); } }
```

Session 21 — JUnit Tests with AEM Mocks

AEM Mocks creates a fake in-memory AEM environment so you can test Sling Models without a running AEM server.

```
@ExtendWith(AemContextExtension.class) class HeroBannerModelTest { private final AemContext
ctx = new AemContext(); @Test void testModelReadsProperties() {
```

```
ctx.load().json("/testdata/hero-page.json", "/content/aembank/home");
ctx.currentResource("/content/aembank/home/jcr:content"); HeroBannerModel model =
ctx.request().adaptTo(HeroBannerModel.class); assertNotNull(model); assertEquals("Welcome",
model.getBannerTitle()); } }
```

Session 22 — Replication & MSM

Concept	What it does
Default Agent	Sends content Author→Publish when editor clicks Publish
Flush Agent	Tells Dispatcher to clear cached HTML after publishing
Reverse Replication	Moves user-submitted content Publish→Author
Blueprint	Master site — source for all Live Copies
Live Copy	Child site connected to Blueprint, inherits content via rollout
Rollout	Pushes Blueprint changes to all Live Copies
Cancelled Inheritance	Property in Live Copy overridden locally, rollout won't touch it
referencesUpdate handler	Fixes internal links in Live Copies to point to Live Copy paths (not Blueprint)

4. Custom Hero Banner Component (Sessions 4–7)

The first custom component built in the project. Covers component structure, Touch UI dialog, HTL, and Sling Model.

4.1 Files Created

File	Location	Purpose
.content.xml	ui.apps/.../components/herobanner/	Registers as cq:Component
_cq_dialog/.content.xml	ui.apps/.../herobanner/_cq_dialog/	Touch UI dialog — 5 fields
HeroBannerModel.java	core/.../models/	Sling Model — reads JCR properties
herobanner.html	ui.apps/.../herobanner/	HTL template
clientlib-herobanner/	ui.apps/.../clientlibs/	CSS + JS for background image

4.2 Component .content.xml — Key Learning

jcr:primaryType must be cq:Component (not nt:folder) for the dialog to work:

```
<jcr:root xmlns:cq="http://www.day.com/jcr/cq/1.0" xmlns:jcr="http://www.jcp.org/jcr/1.0"
jcr:primaryType="cq:Component" jcr:title="Hero Banner" componentGroup="AEM Bank Website -
Content"/>
```

4.3 HTL Template — Key Learning

Use data-sly-test for conditional rendering. Never use dynamic style attributes — use data-bg instead:

```
<sly data-sly-use.model="com.aembank.core.models.HeroBannerModel"/> <div class="herobanner"
data-bg="${model.backgroundImage}" data-sly-test="${model.bannerTitle}">
<h1>${model.bannerTitle}</h1> <p
data-sly-test="${model.bannerSubtitle}">${model.bannerSubtitle}</p> <a
href="${model.ctaLink}" data-sly-test="${model.hasCtaButton}"> ${model.ctaText} </a> </div>
```

5. Daily Developer Tools & URLs

URL	Purpose
localhost:4502/sites.html	Sites console — browse and edit pages
localhost:4502/crx/de	CRXDE Lite — browse JCR repository
localhost:4502/system/console/bundles	Felix Console — check bundle status (Active = green)
localhost:4502/system/console/components	Felix Console — check OSGi component status
localhost:4502/system/console/configMgr	Felix Console — manage OSGi configurations
localhost:4502/system/console/slinglog	Felix Console — manage log files
localhost:4502/system/console/status-slingmodels	Felix Console — see registered Sling Models
localhost:4502/libs/cq/tagging/gui/content/tags	Tagging console — create/manage tags
localhost:4502/assets.html/content/dam	Assets console — upload and manage DAM assets
localhost:4502/libs/wcm/core/content/sites/templates/conf/aembank	Template editor
localhost:4502/libs/cq/search/content/querydebug	QueryBuilder debugger
localhost:4502/crx/packmgr	Package Manager — upload/download content packages
localhost:4502/etc/replication/agents.author.html	Replication agents console
localhost:4502/workflows.html	Workflow console

5.1 Common Errors & Fixes

Error	Cause	Fix
handshake_failure	Corporate SSL proxy blocks HTTPS	Add -Dhttps.protocols=TLSv1.2 to Maven command
No POM in directory	Running archetype:generate without port	Create temp pom.xml with <packaging>pom</packaging>
class X should be in file WrongClass.java	Wrong class placed into wrong file	Create a new separate .java file for each public class
cq:dialog not allowed as component content	Component content file missing or wrong primary type	Set primary type='cq:Component' in herobanner/.content.xml
HTL style attribute warning	Dynamic value in style= attribute	Use data-bg attribute, apply via JavaScript
Bundle in Installed state	Missing OSGi dependency or Java error	Check Felix Console → bundle → error message
adaptTo returns null	Missing @Model annotation or wrong adapter	Apply @Model annotation and OPTIONAL injection strategy

6. Project Roadmap — Custom Features To Build

■ Phase 1 — DONE

- Project generated with Maven Archetype 56
- Deployed successfully to AEM 6.5
- Hero Banner component — .content.xml, dialog, Sling Model, HTL, CSS/JS clientlib

■ Phase 2 — Components (Sessions 3–8)

- Loan Product Card component with multifield dialog
- Bank Page Template with locked header/footer and content policies
- Tag List component using TagManager API

■ Phase 3 — Backend Services (Sessions 13–17)

- BankConfigService — OSGi service with configuration factory
- ProductSearchServlet — Sling servlet returning JSON results
- Service user 'bankwebsite-service' with repoint permissions
- Custom log file bankwebsite.log with runmode-based configs
- Scheduled job reading /content/aembank pages using service user

■ Phase 4 — Advanced (Sessions 18–22)

- Tag namespace 'bankproducts' with loans, cards, savings, insurance tags
- Custom metadata schema for DAM assets (Product SKU, Season, Product Type)
- ValidateAndPublishProcess workflow step
- JUnit tests with AEM Mocks + Mockito for all models
- Replication agent configuration + PublishPageServlet
- Blueprint + Live Copies for India and Singapore sites